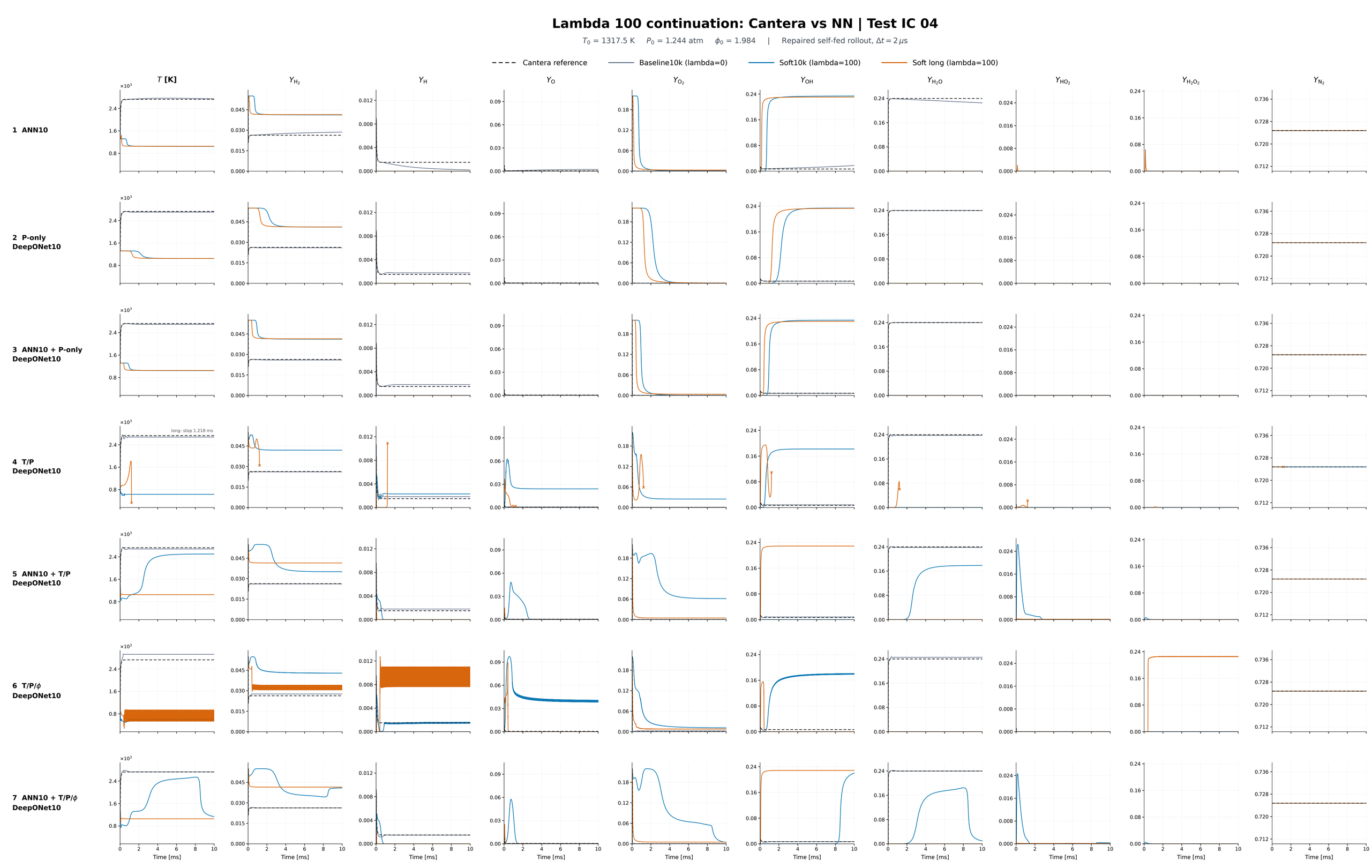
Lambda 100 continuation: Cantera vs NN | Test IC 04

T_0 = 1317.5 K P_0 = 1.244 atm ϕ_0 = 1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 04 | Cantera vs NN NRMSE

T0=1317.5 K P0=1.244 atm phi0=1.984 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0108	0.0583	0.593	2.33	0.0953	0.832	0.0354	0.149	0.418	0	Complete
1 ANN10	soft10k	0.608	0.621	1	1	5.54	31.4	1	1.07	109	0	Complete
1 ANN10	long	0.612	0.584	1	1	2.34	32.1	1	35.6	3.07e+04	0	Complete
2 P-only DeepONet10	base10k	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	soft10k	0.593	0.718	1	1	9.71	28.3	1	1.02	3.14	0	Complete
2 P-only DeepONet10	long	0.602	0.66	1	1	7.39	30.2	1	0.997	1.02	0	Complete
3 ANN10 + P-only DeepONet10	base10k	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.606	0.637	1	1	6.4	31	1	1.07	34	0	Complete
3 ANN10 + P-only DeepONet10	long	0.609	0.606	1	1	4.22	31.6	1	0.988	86	0	Complete
4 T/P DeepONet10	base10k	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	soft10k	0.767	0.628	0.526	46.2	4.7	24.4	1	10.1	7.44	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 1.218 ms
5 ANN10 + T/P DeepONet10	base10k	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.318	0.618	0.973	23.7	10.9	1	0.56	1.53e+03	6.86e+03	0	Complete
5 ANN10 + T/P DeepONet10	long	0.613	0.579	0.982	1.49	1.28	32.1	1	45.5	335	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.779	0.691	0.307	80	4.51	23	1	12	5.86	0	Complete
6 T/P/ϕ DeepONet10	long	0.743	0.273	5.13	16.4	1.31	3.92	1	6.59	1.74e+06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.384	0.675	0.99	20.7	11.9	10.7	0.682	1.36e+03	5.05e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.613	0.581	0.987	2.47	1.34	32	1	43.2	171	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 05

T_0 = 1049.0 K P_0 = 4.666 atm ϕ_0 = 0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

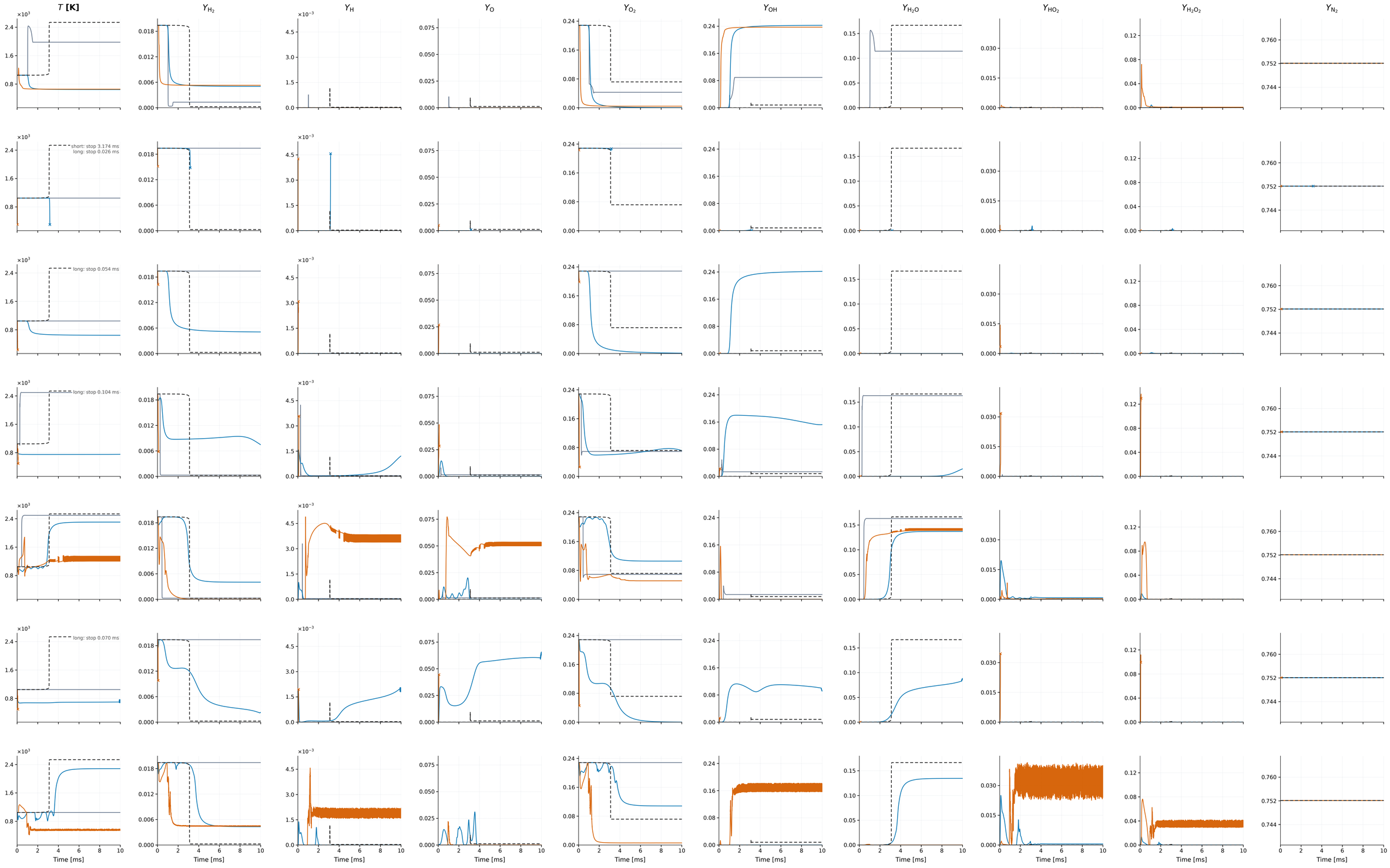
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 05 | Cantera vs NN NRMSE

T0=1049.0 K P0=4.666 atm phi0=0.674 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.299	0.773	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	soft10k	0.722	0.669	1	1	0.798	32.4	1	0.961	3.53	0	Complete
1 ANN10	long	0.72	0.797	1	1	0.939	33.7	1	2.31	37.9	0	Complete
2 P-only DeepONet10	base10k	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 3.174 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
3 ANN10 + P-only DeepONet10	base10k	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.72	0.641	1	1	0.746	31.6	1	0.954	1.57	0	Complete
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.054 ms
4 T/P DeepONet10	base10k	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	soft10k	0.682	0.831	12.4	2.5	0.593	23.5	0.987	1	1	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.104 ms
5 ANN10 + T/P DeepONet10	base10k	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.106	0.336	3.97	3.99	0.236	1	0.21	55.1	7.6	0	Complete
5 ANN10 + T/P DeepONet10	long	0.488	0.836	111	48.1	0.608	2.22	0.475	9.13	116	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.706	0.481	31.8	47.7	0.553	14	0.608	1	1	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.2	0.523	5.71	6.17	0.331	1	0.342	68.2	9.82	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.758	0.679	54.6	2.14	0.781	22.6	1	491	222	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 07

T_0 = 930.6 K P_0 = 3.722 atm ϕ_0 = 0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=100) Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

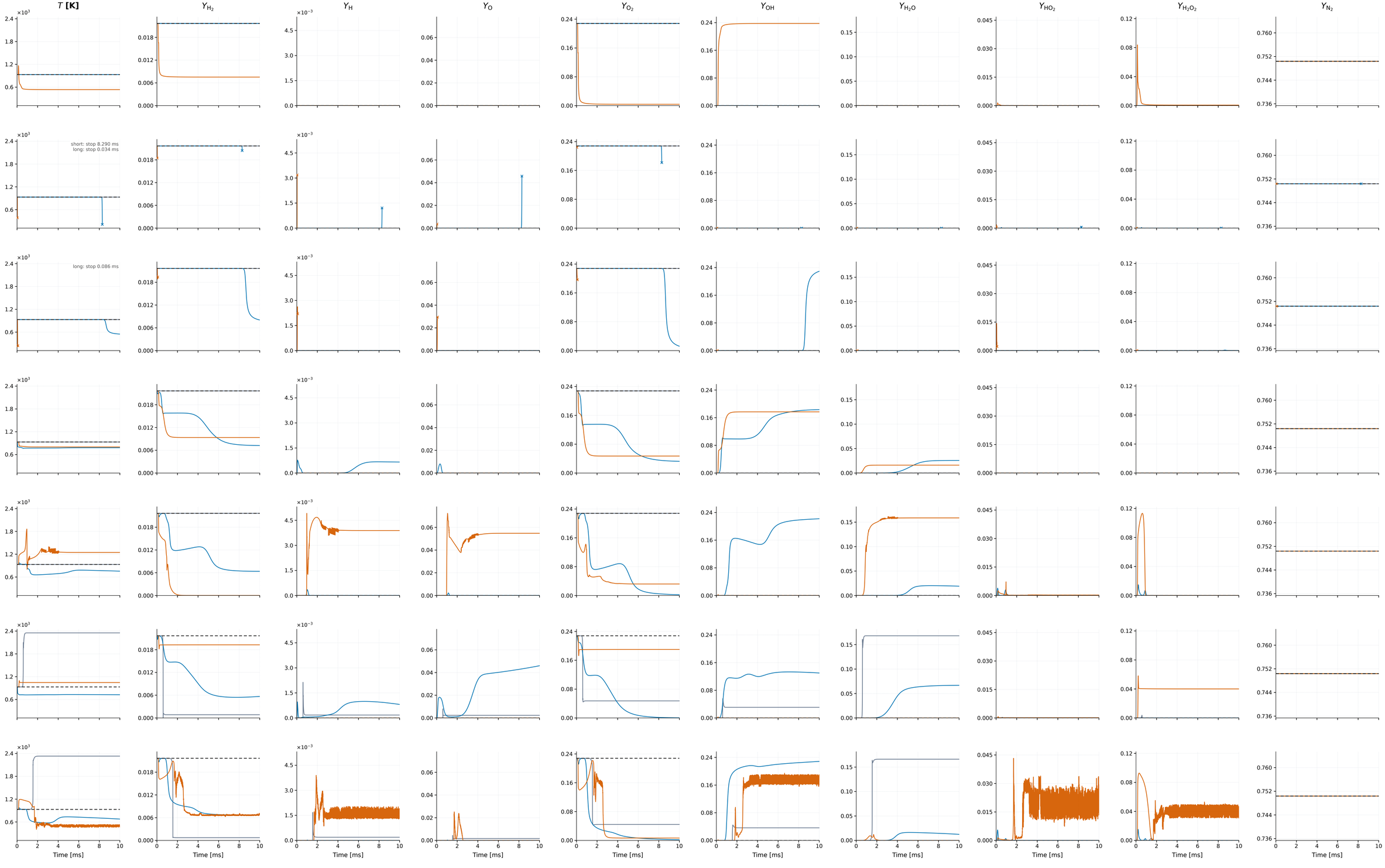
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 07 | Cantera vs NN NRMSE

T0=930.6 K P0=3.722 atm phi0=0.754 | Repaired self-fed,10ms

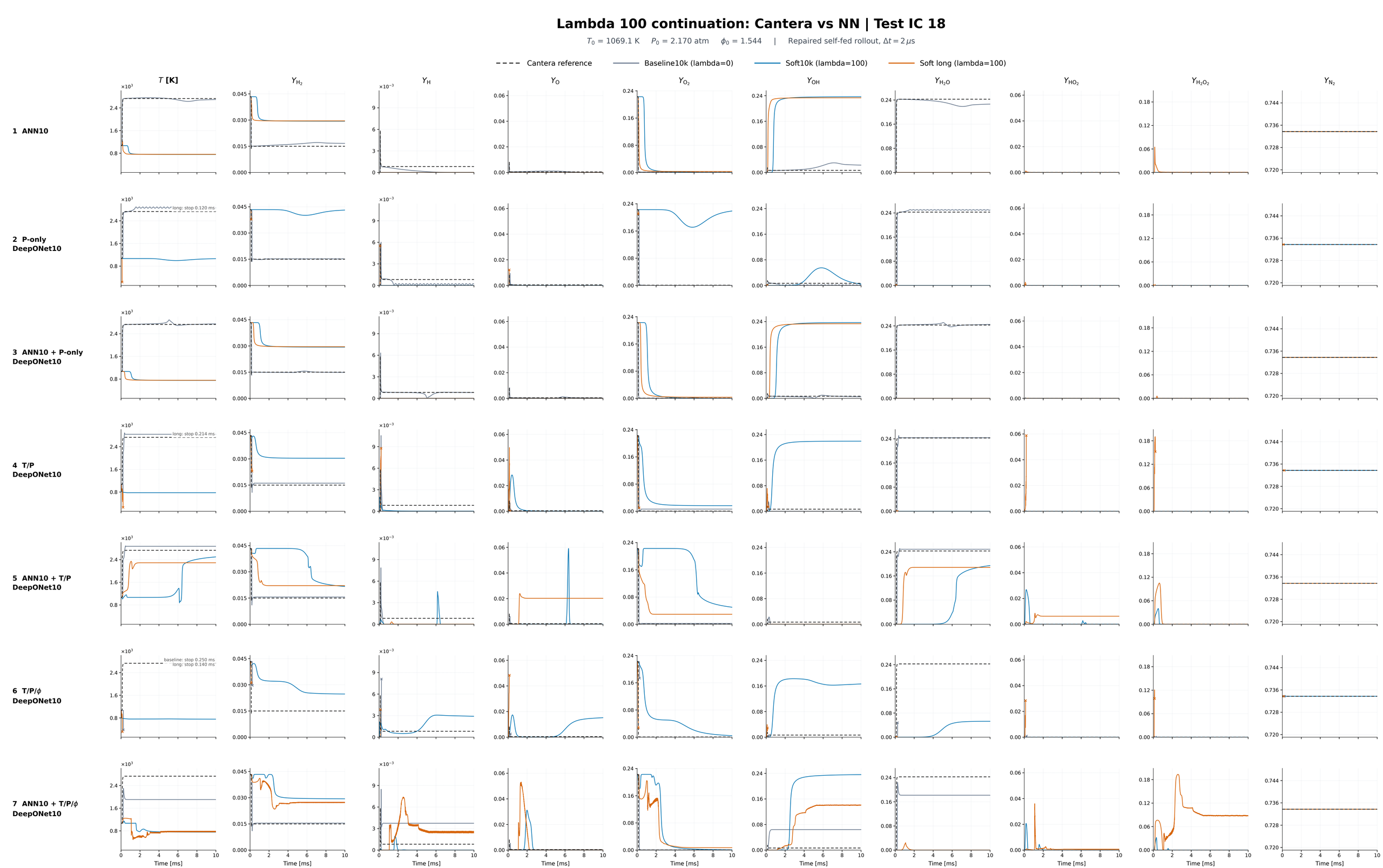
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	**soft10k**	2.84e-06	2.61e-06	1	51.2	4.32e-06	1	1	0.999	1	0	Complete
1 ANN10	**long**	0.416	0.644	150	8.63	0.971	5.18e+10	1	284	1.63e+04	0	Complete
2 P-only DeepONet10	**base10k**	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 8.290 ms
2 P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.034 ms
3 ANN10 + P-only DeepONet10	**base10k**	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.136	0.209	26.1	409	0.315	1.69e+10	1	39.1	272	0	Complete
3 ANN10 + P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.086 ms
4 T/P DeepONet10	**base10k**	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	**soft10k**	0.164	0.487	1.21e+08	1.62e+08	0.65	3.18e+10	9.96e+04	1	1	0	Complete
4 T/P DeepONet10	**long**	0.137	0.546	1.48e+04	0.636	0.763	3.78e+10	9.28e+04	7.29	1	0	Complete
5 ANN10 + T/P DeepONet10	**base10k**	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.207	0.548	1.07e+07	3.08e+07	0.78	3.95e+10	8.07e+04	1.02e+03	4.11e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.344	0.944	1.05e+09	6.77e+09	0.808	3.26e+07	8.93e+05	1.03e+03	6.72e+04	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.218	0.606	1.99e+08	4.47e+09	0.819	2.70e+10	3.01e+05	1	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	0.123	0.109	1	2.46e+03	0.165	4.47e+07	0.893	253	1.01e+05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	1.38	0.888	5.23e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.264	0.613	4.48e+05	3.08e+06	0.885	4.54e+10	6.99e+04	1.04e+03	3.59e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.427	0.602	4.32e+08	5.08e+08	0.845	3.34e+10	1.33e+04	3.68e+04	1.13e+05	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 18

T_0 = 1069.1 K P_0 = 2.170 atm ϕ_0 = 1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 18 | Cantera vs NN NRMSE

T0=1069.1 K P0=2.170 atm phi0=1.544 | Repaired self-fed,10ms

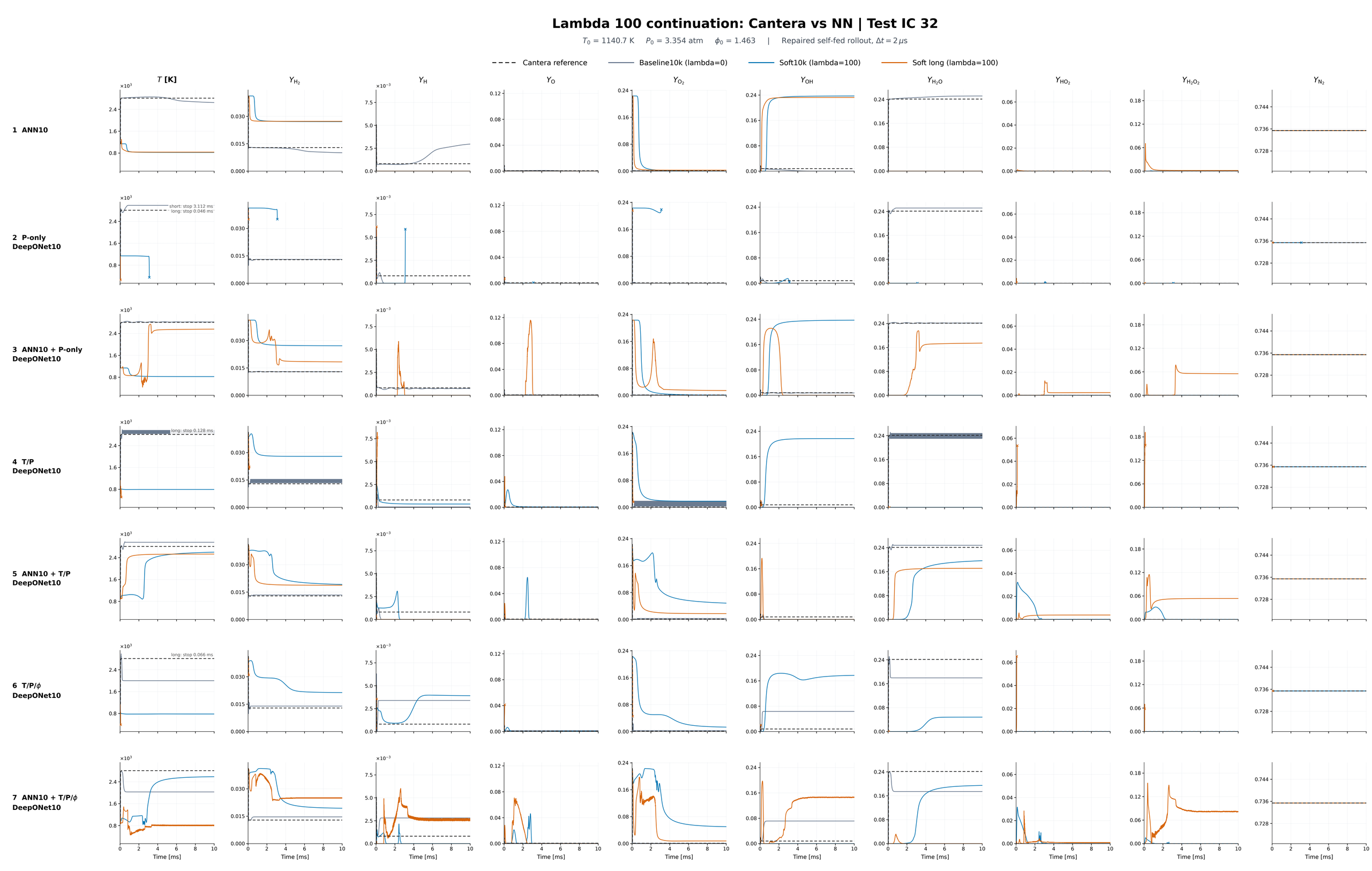
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	**soft10k**	0.714	1	1	1	2.24	32.8	1	1.04	58.6	0	Complete
1 ANN10	**long**	0.719	0.931	1	1	0.537	33.5	1	7.29	4.09e+03	0	Complete
2 P-only DeepONet10	**base10k**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**soft10k**	0.617	1.73	1	1	8.16	3.44	1	1.05	1.07	0	Complete
2 P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.120 ms
3 ANN10 + P-only DeepONet10	**base10k**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.711	1.04	1	1	2.72	32.1	1	1.03	15.8	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.717	0.965	1	1	1.44	33	1	1.31	402	0	Complete
4 T/P DeepONet10	**base10k**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**soft10k**	0.714	1.04	0.96	10.9	1.91	30.3	1	1.02	1	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.214 ms
5 ANN10 + T/P DeepONet10	**base10k**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.484	1.42	1.06	13.3	6.91	1	0.795	331	5.83e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.209	0.59	0.989	37.5	1.9	0.999	0.343	418	2.02e+04	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	—	—	—	—	—	—	—	—	—	—	Stop 0.250 ms
6 T/P/ϕ DeepONet10	**soft10k**	0.72	0.936	1.83	17.6	2.47	23.7	0.897	1.03	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.702	1.19	0.99	12.7	4.1	29	1	187	3.31e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.715	0.967	2.77	22.2	2.89	16.3	0.996	165	8.15e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 32

T_0 = 1140.7 K P_0 = 3.354 atm ϕ_0 = 1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 32 | Cantera vs NN NRMSE

T0=1140.7 K P0=3.354 atm phi0=1.463 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	soft10k	0.698	1.19	1	1	4.15	26.9	1	1.26	156	0	Complete
1 ANN10	long	0.699	1.12	1	1	1.68	27.2	1	13.6	7.57e+03	0	Complete
2 P-only DeepONet10	base10k	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 3.112 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
3 ANN10 + P-only DeepONet10	base10k	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.694	1.24	1	1	4.87	26.3	1	1.17	43.5	0	Complete
3 ANN10 + P-only DeepONet10	long	0.381	0.842	1.2	36.2	3.73	9.94	0.557	241	5.42e+04	0	Complete
4 T/P DeepONet10	base10k	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	soft10k	0.716	1.22	0.553	7.62	3.7	24.6	1	0.599	0.893	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.128 ms
5 ANN10 + T/P DeepONet10	base10k	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.338	1.07	1.02	13.4	7.48	1	0.534	910	1.28e+04	0	Complete
5 ANN10 + T/P DeepONet10	long	0.174	0.605	0.996	2.83	2.52	2.78	0.38	326	6.39e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.721	1.01	2.96	1.62	4.44	19.8	0.883	0.381	0.845	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.066 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.353	1.24	0.948	11.3	8.95	1	0.578	540	2.88e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.718	1.16	2.46	28.3	4.86	14.4	0.996	215	9.45e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 39

T_0 = 1250.1 K P_0 = 2.297 atm ϕ_0 = 0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

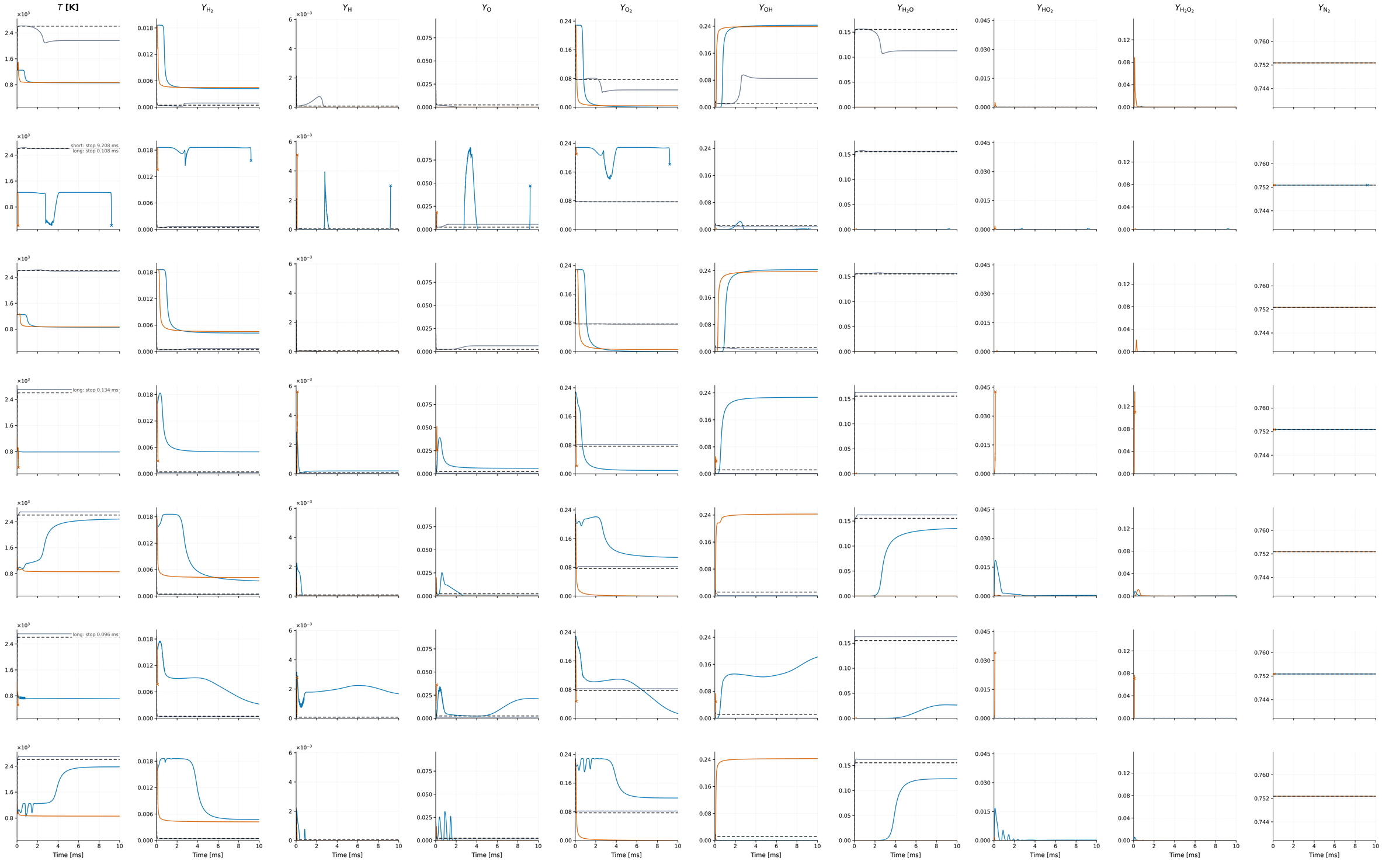
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 39 | Cantera vs NN NRMSE

T0=1250.1 K P0=2.297 atm phi0=0.646 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	soft10k	0.658	7.05	1	1	1.05	18.6	1	0.909	214	0	Complete
1 ANN10	long	0.664	5.02	1	1	0.938	19	1	8.48	5.93e+03	0	Complete
2 P-only DeepONet10	base10k	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 9.208 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
3 ANN10 + P-only DeepONet10	base10k	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.654	7.79	1	1	1.07	18.2	1	1.05	103	0	Complete
3 ANN10 + P-only DeepONet10	long	0.66	5.91	1	1	0.929	18.6	1	2.66	1.52e+03	0	Complete
4 T/P DeepONet10	base10k	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	soft10k	0.699	7.11	3.38	3.36	0.876	17.2	1	1.44	1.16	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.134 ms
5 ANN10 + T/P DeepONet10	base10k	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.303	11.1	3.75	1.97	1.02	1	0.541	191	1.18e+03	0	Complete
5 ANN10 + T/P DeepONet10	long	0.668	4.69	0.997	1.06	0.969	19.3	1	4.56	1.86e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.724	9.06	20.3	4.6	0.563	10.4	0.936	1.16	1.02	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.096 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.345	13.2	2.4	2.25	1.25	1	0.643	134	619	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.668	4.73	1.23	1.03	0.971	19.3	1	5.38	168	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 44

T_0 = 1031.1 K P_0 = 2.272 atm ϕ_0 = 0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

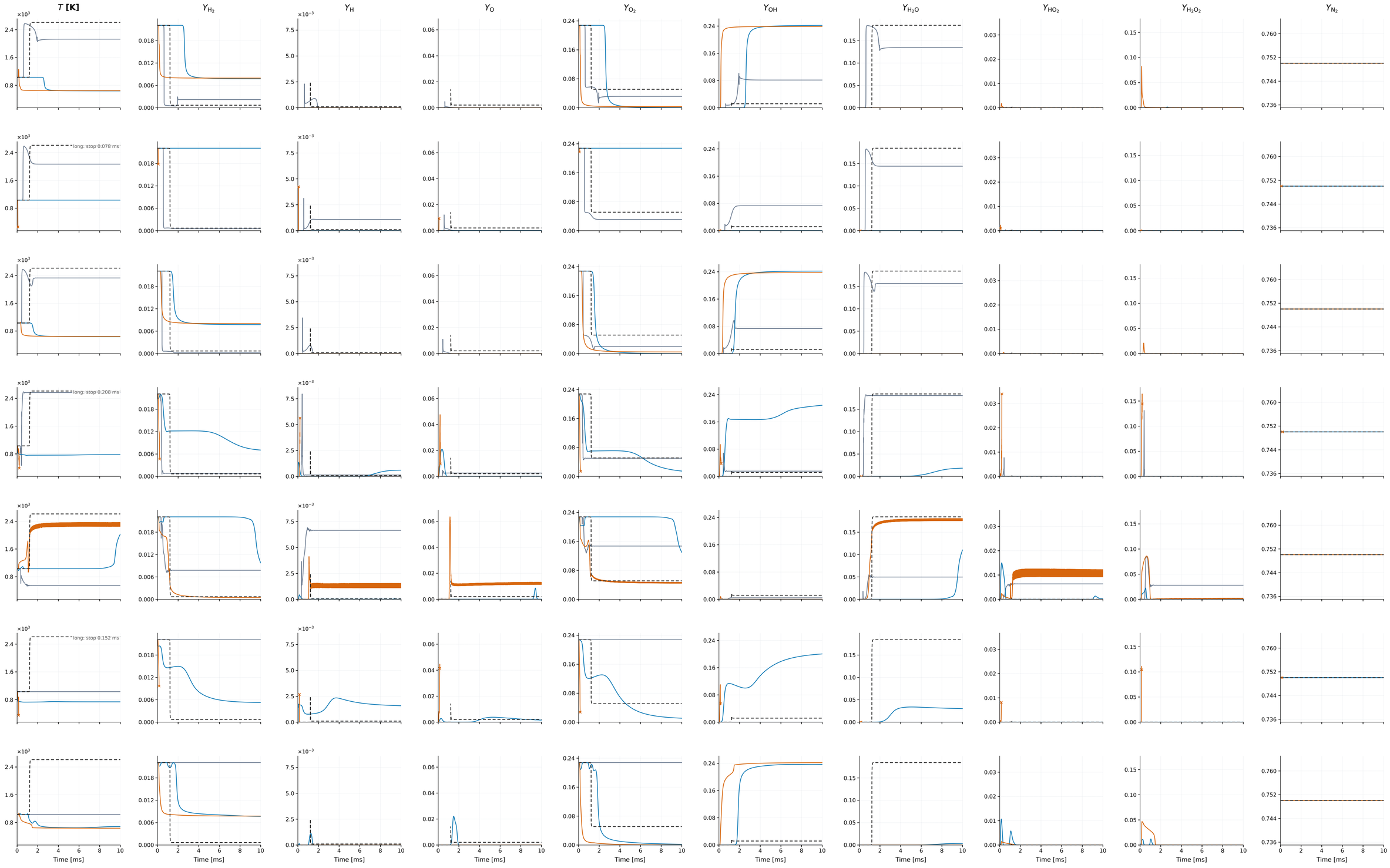
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 44 | Cantera vs NN NRMSE

T0=1031.1 K P0=2.272 atm phi0=0.770 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	soft10k	0.724	1.32	1	1	0.832	17.5	1	0.984	26.5	0	Complete
1 ANN10	long	0.747	1.05	1	1	0.891	20.3	1	4.61	682	0	Complete
2 P-only DeepONet10	base10k	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	soft10k	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.078 ms
3 ANN10 + P-only DeepONet10	base10k	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.74	1.01	1	1	0.565	18.7	1	0.963	7.61	0	Complete
3 ANN10 + P-only DeepONet10	long	0.745	1.02	1	1	0.781	19.8	1	1.7	188	0	Complete
4 T/P DeepONet10	base10k	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	soft10k	0.698	1.24	2.51	2.13	0.466	15	0.968	1	1	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.208 ms
5 ANN10 + T/P DeepONet10	base10k	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.582	2.53	1.09	1.04	1.74	1	0.972	88.6	407	0	Complete
5 ANN10 + T/P DeepONet10	long	0.123	0.326	10.3	5.39	0.315	1	0.105	352	2.59e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.711	1.03	14.6	0.705	0.539	13.1	0.863	1	1	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.152 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.732	1.13	1.55	1.97	0.595	17.9	0.995	51.2	239	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.746	1.02	1	1	0.873	20.1	1	9.71	1.46e+03	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 45

T_0 = 1181.9 K P_0 = 4.842 atm ϕ_0 = 1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Columns: T [K] · Y_{H_2} · Y_H · Y_O · Y_{O_2} · Y_{OH} · Y_{H_2O} · Y_{HO_2} · $Y_{H_2O_2}$ · Y_{N_2}

Row 2: short: stop 1.658 ms; long: stop 0.022 ms

Row 4: long: stop 0.068 ms

Row 6: long: stop 0.046 ms

Time [ms]

Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 45 | Cantera vs NN NRMSE

T0=1181.9 K P0=4.842 atm phi0=1.099 | Repaired self-fed,10ms

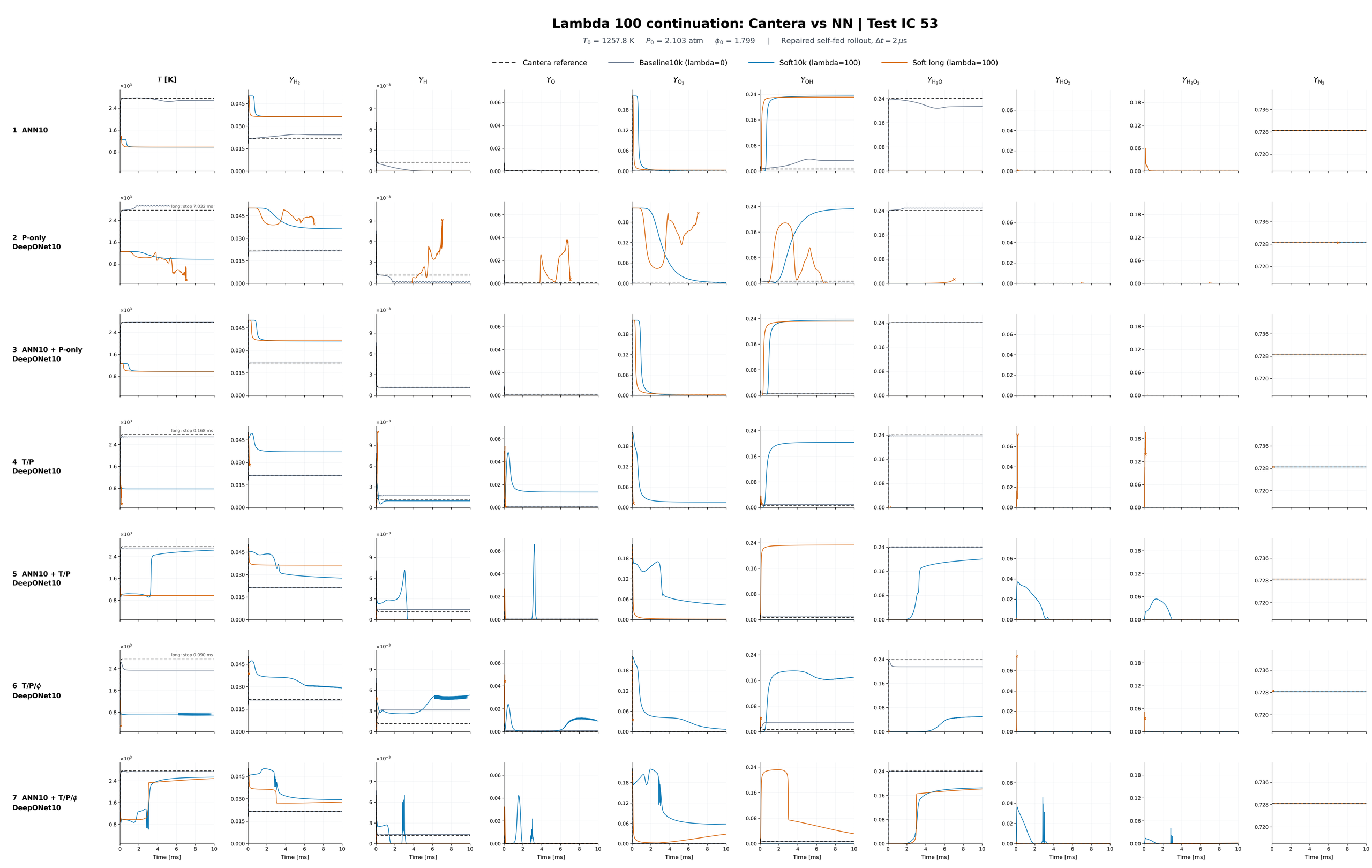
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**base10k**	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	**soft10k**	0.702	2.97	1	1	4.13	14.6	1	1.5	229	0	Complete
1 ANN10	**long**	0.7	2.74	1	1	1.71	14.6	1	13	6.02e+03	0	Complete
2 P-only DeepONet10	**base10k**	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	**soft10k**	—	—	—	—	—	—	—	—	—	—	Stop 1.658 ms
2 P-only DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.022 ms
3 ANN10 + P-only DeepONet10	**base10k**	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	**soft10k**	0.641	2.94	1	1	4.64	13	0.93	1.63	78.9	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.416	2.71	0.965	19.3	5.04	0.985	0.51	462	3.40e+04	0	Complete
4 T/P DeepONet10	**base10k**	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	**soft10k**	0.74	3.2	1.3	1.91	4.36	11.4	1	1.66	0.897	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.068 ms
5 ANN10 + T/P DeepONet10	**base10k**	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	**soft10k**	0.355	3.13	1.08	5.45	8.23	1	0.55	316	1.03e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.169	1.52	0.994	1.81	2.56	1.62	0.361	408	3.63e+04	0	Complete
6 T/P/ϕ DeepONet10	**base10k**	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	**soft10k**	0.755	3.2	1.08	25.7	3.81	7.77	0.856	1.98	0.906	0	Complete
6 T/P/ϕ DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.349	3.05	1.15	5.29	8.24	1	0.543	358	1.15e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.757	4.29	3.3	18.6	8.58	3.48	0.982	456	3.69e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 53

T_0 = 1257.8 K P_0 = 2.103 atm ϕ_0 = 1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 53 | Cantera vs NN NRMSE

T0=1257.8 K P0=2.103 atm phi0=1.799 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	soft10k	0.64	0.74	1	1	5.93	31.9	1	1.18	182	0	Complete
1 ANN10	long	0.644	0.697	1	1	2.62	32.5	1	19.1	2.10e+04	0	Complete
2 P-only DeepONet10	base10k	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	soft10k	0.613	0.933	1	1	12.3	25.4	1	1.18	3.84	0	Complete
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 7.032 ms
3 ANN10 + P-only DeepONet10	base10k	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.638	0.763	1	1	6.98	31.3	1	1.17	62.5	0	Complete
3 ANN10 + P-only DeepONet10	long	0.642	0.72	1	1	4.34	32.1	1	1.11	417	0	Complete
4 T/P DeepONet10	base10k	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	soft10k	0.719	0.769	0.306	32.9	5.27	27.4	1	4.46	3.18	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.168 ms
5 ANN10 + T/P DeepONet10	base10k	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.366	0.63	1.39	15.4	10.4	1	0.571	2.87e+03	1.04e+05	0	Complete
5 ANN10 + T/P DeepONet10	long	0.646	0.688	0.99	3.53	1.73	32.8	1	17.3	2.07e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.742	0.621	2.39	15.2	6.76	23.7	0.915	6.91	4.35	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.090 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.352	0.746	1.09	15.1	12.8	1	0.587	1.8e+03	1.82e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.364	0.447	0.991	4.56	2.32	18.3	0.588	32.9	5.96e+03	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 57

T_0 = 1412.0 K P_0 = 4.083 atm ϕ_0 = 0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

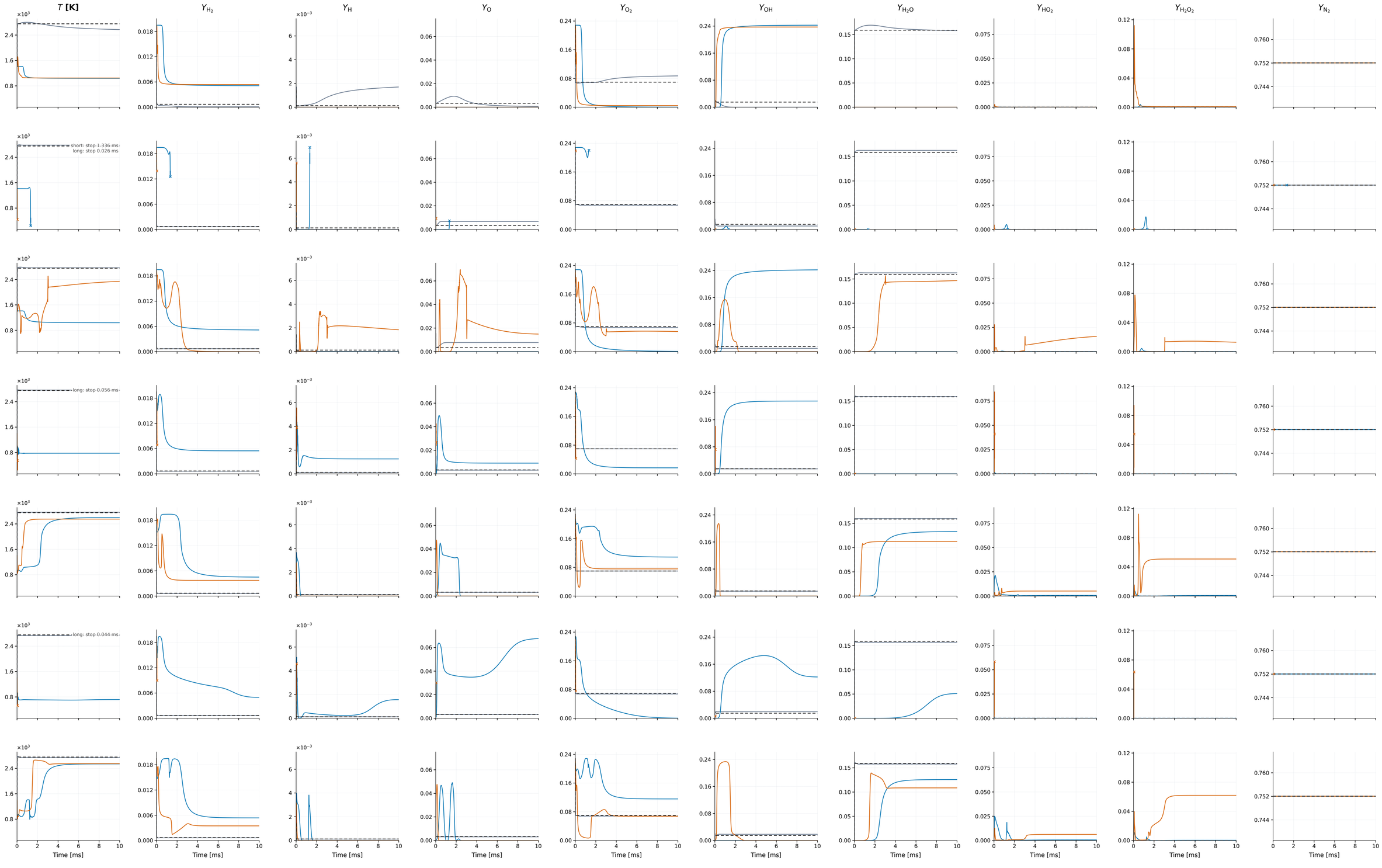
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 57 | Cantera vs NN NRMSE

T0=1412.0 K P0=4.083 atm phi0=0.677 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	soft10k	0.613	7.97	1	1	1.08	14.4	1	1.09	269	0	Complete
1 ANN10	long	0.616	6.11	1	1	0.929	14.6	1	7.73	5.1e+03	0	Complete
2 P-only DeepONet10	base10k	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.336 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.026 ms
3 ANN10 + P-only DeepONet10	base10k	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.609	8.62	1	1	1.06	14	1	3.33	362	0	Complete
3 ANN10 + P-only DeepONet10	long	0.335	7.85	13.9	6.53	0.555	2.81	0.477	379	8.52e+03	0	Complete
4 T/P DeepONet10	base10k	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	soft10k	0.718	7.91	9.33	3.15	0.807	12.5	1	5.59	4.87	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.056 ms
5 ANN10 + T/P DeepONet10	base10k	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.308	11.7	3.88	4.23	1.02	1	0.512	127	610	0	Complete
5 ANN10 + T/P DeepONet10	long	0.162	5.19	1.34	1.57	0.297	2.43	0.379	182	2.95e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.742	10.2	6.01	13.8	0.844	9.14	0.907	2.5	2.49	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.044 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.307	12.1	5.05	3.52	1.15	1	0.537	163	789	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.246	4.68	1.6	1.76	0.358	5.08	0.469	194	3.14e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 64

T_0 = 1045.3 K P_0 = 4.301 atm ϕ_0 = 0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

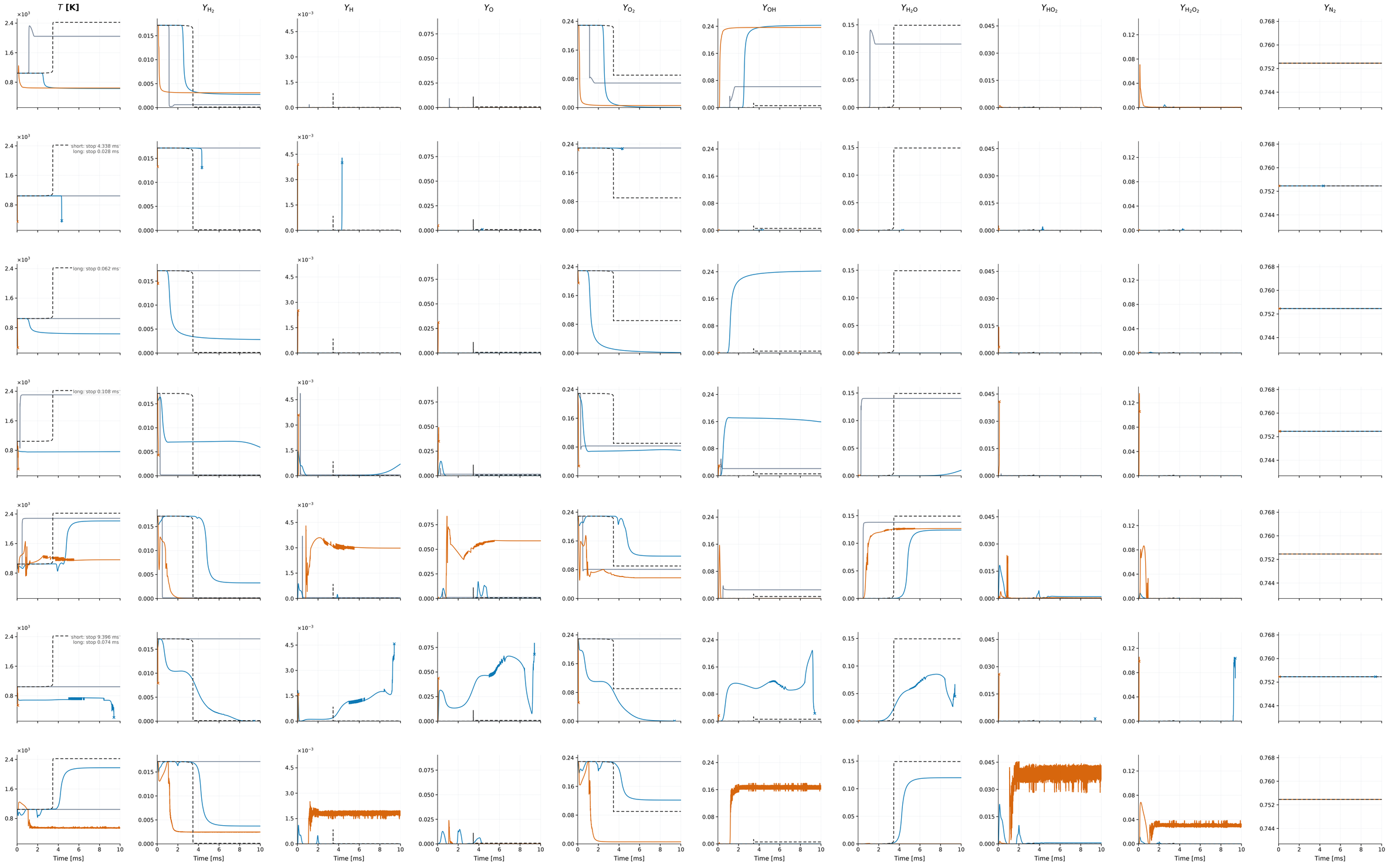
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 64 | Cantera vs NN NRMSE

T0=1045.3 K P0=4.301 atm phi0=0.595 | Repaired self-fed,10ms

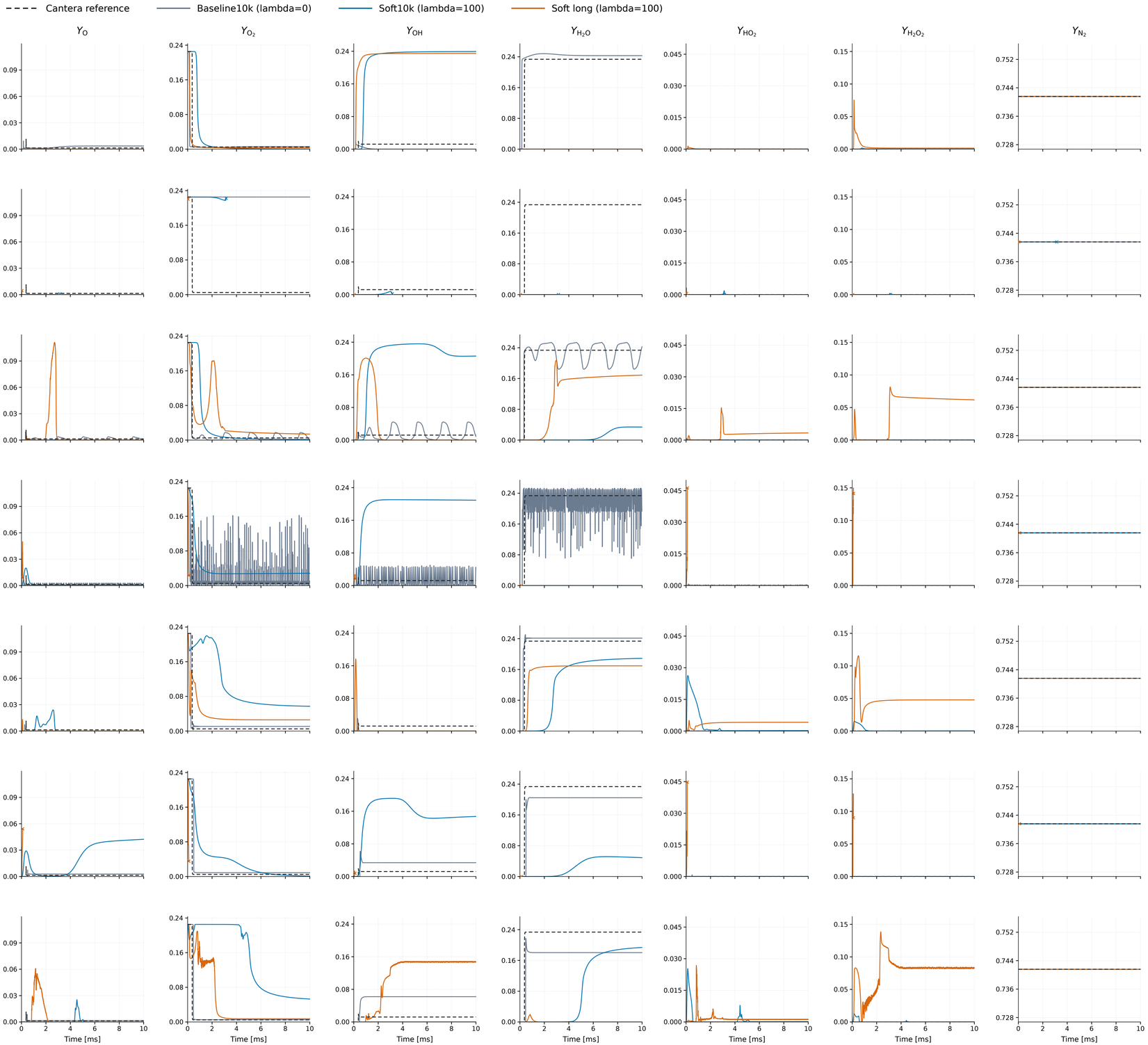
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.284	0.791	1.02	1.04	0.509	11.4	0.523	1.02	1.01	0	Complete
1 ANN10	soft10k	0.706	0.434	1	1	0.608	43.1	1	0.762	3.48	0	Complete
1 ANN10	long	0.709	0.822	1	1	0.937	49.2	1	2.15	36.4	0	Complete
2 P-only DeepONet10	base10k	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 4.338 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.028 ms
3 ANN10 + P-only DeepONet10	base10k	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.708	0.634	1	1	0.768	45.9	1	0.98	1.66	0	Complete
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.062 ms
4 T/P DeepONet10	base10k	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	soft10k	0.659	0.769	14.6	3.73	0.567	34.1	0.989	1	1	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.108 ms
5 ANN10 + T/P DeepONet10	base10k	0.331	0.919	4.92	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.258	0.638	6.14	4.8	0.333	1	0.462	54.8	7.83	0	Complete
5 ANN10 + T/P DeepONet10	long	0.498	0.846	158	79.8	0.569	3.18	0.523	23	122	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 9.396 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.074 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.208	0.536	7.08	5.16	0.29	1	0.383	60.5	8.64	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.748	0.702	91.8	2.72	0.81	32.6	1	611	212	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 68

T_0 = 1086.3 K P_0 = 3.702 atm ϕ_0 = 1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 68 | Cantera vs NN NRMSE

T0=1086.3 K P0=3.702 atm phi0=1.168 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

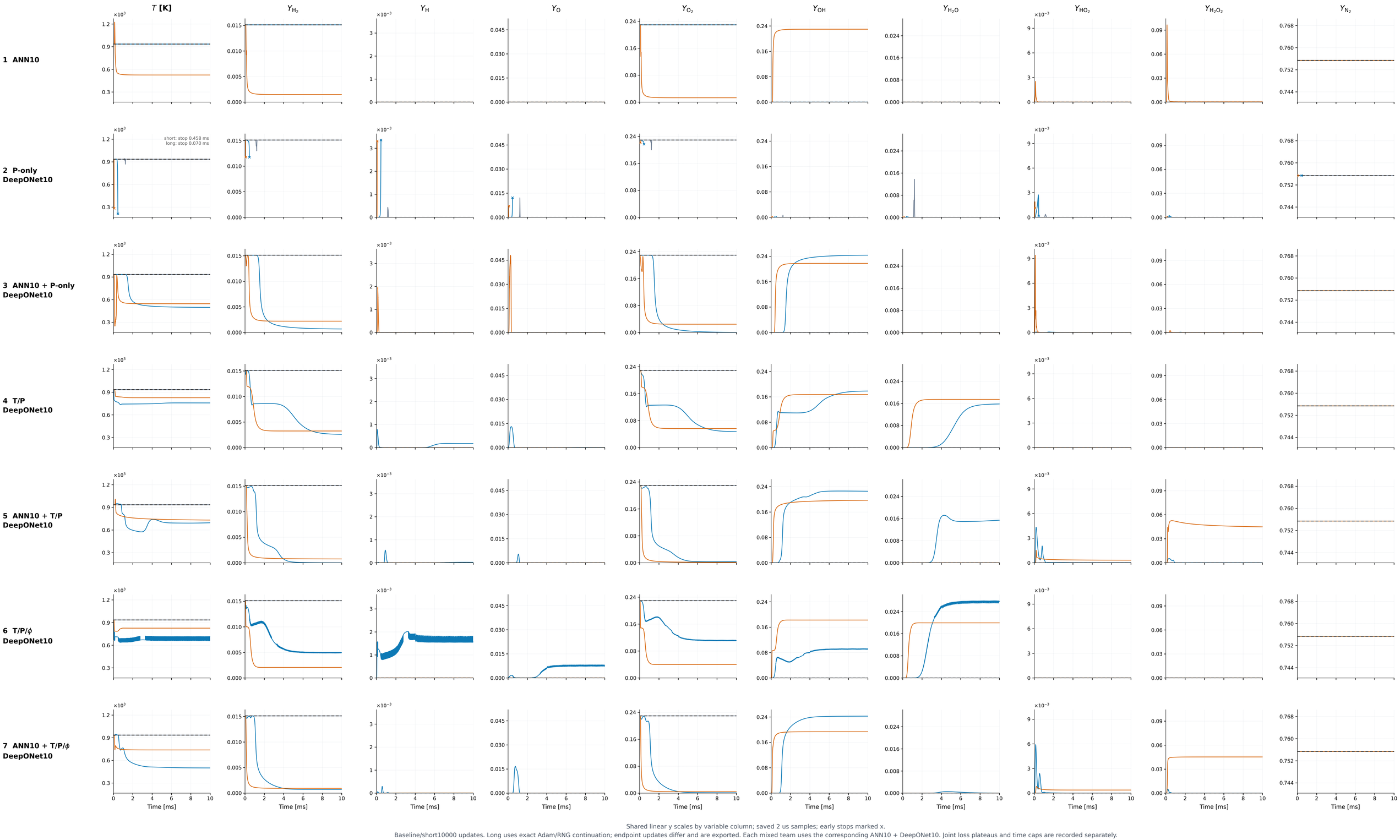
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	soft10k	0.73	1.64	1	1	1	17.9	1	1.11	28.6	0	Complete
1 ANN10	long	0.732	1.56	1	1	0.648	18.3	1	6.32	796	0	Complete
2 P-only DeepONet10	base10k	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 3.150 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.036 ms
3 ANN10 + P-only DeepONet10	base10k	0.0874	0.309	0.756	0.913	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.69	1.66	1	1	1.27	16.6	0.957	1.07	7.29	0	Complete
3 ANN10 + P-only DeepONet10	long	0.367	1.2	0.974	15.8	1.07	5.7	0.534	134	6.35e+03	0	Complete
4 T/P DeepONet10	base10k	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	soft10k	0.722	1.77	1.06	2.5	0.793	15.8	1	1.03	1	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.106 ms
5 ANN10 + T/P DeepONet10	base10k	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.329	1.66	0.997	3.56	2.51	1	0.524	254	400	0	Complete
5 ANN10 + T/P DeepONet10	long	0.134	0.783	1	1.11	0.88	1.77	0.321	163	5.86e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.73	1.64	0.96	20.4	0.919	12.2	0.881	1.02	1	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.098 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.438	2.27	1	2.46	3.53	1	0.714	152	238	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.722	1.65	3.19	9.21	1.46	9.62	0.997	99.7	9.47e+03	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 69

T_0 = 935.0 K P_0 = 2.045 atm ϕ_0 = 0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 69 | Cantera vs NN NRMSE

T0=935.0 K P0=2.045 atm phi0=0.523 | Repaired self-fed,10ms

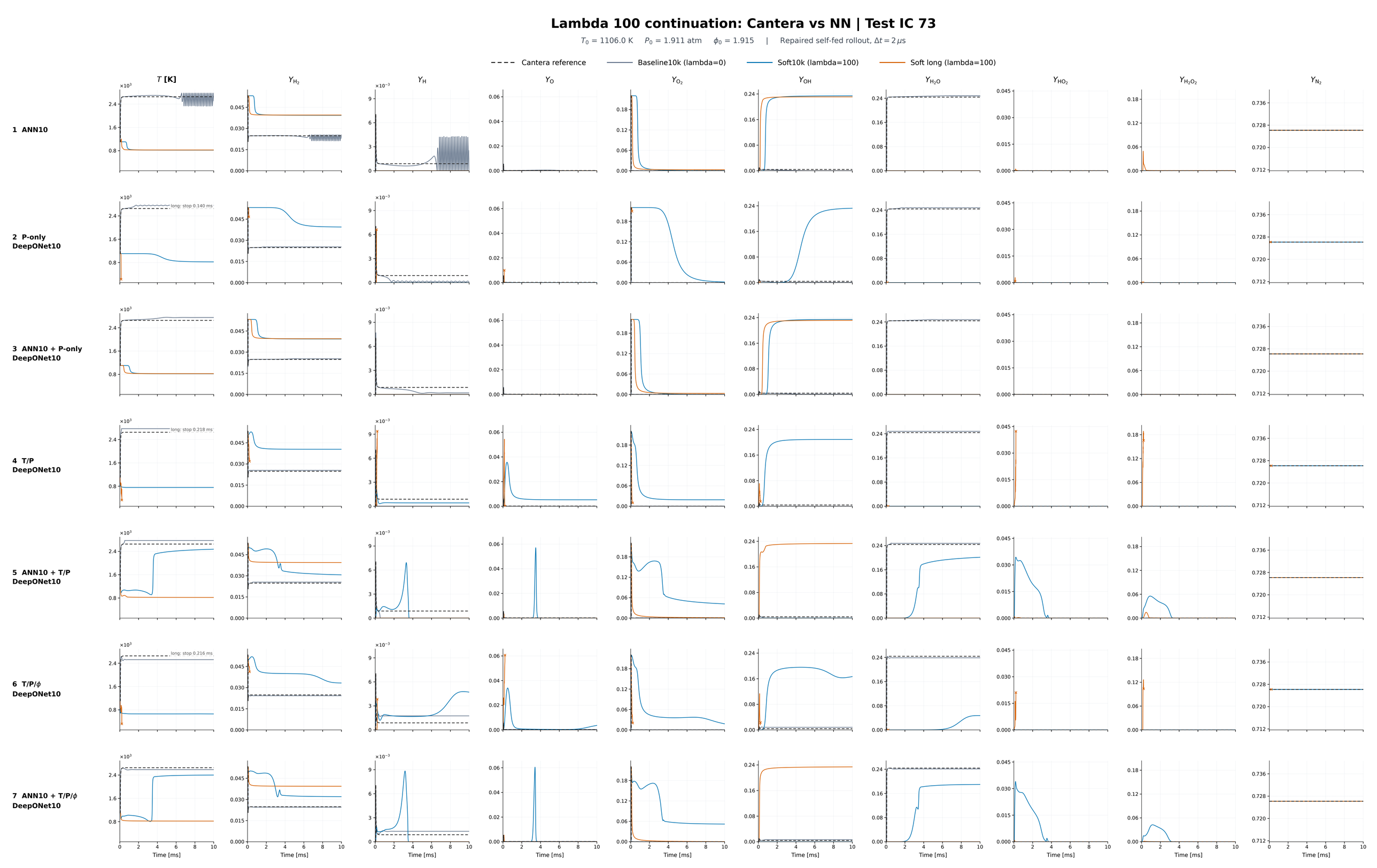
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	soft10k	9.75e-07	1.29e-06	9.98	87.1	1.56e-06	1	1	0.998	1	0	Complete
1 ANN10	long	0.432	0.891	3.27e+03	38.4	0.932	6.36e+10	1	741	7.34e+04	0	Complete
2 P-only DeepONet10	base10k	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 0.458 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
3 ANN10 + P-only DeepONet10	base10k	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.413	0.846	227	67	0.885	6.06e+10	1	69.2	746	0	Complete
3 ANN10 + P-only DeepONet10	long	0.413	0.829	4.79e+07	5.92e+08	0.867	5.93e+10	1	2.74e+03	2.61e+03	0	Complete
4 T/P DeepONet10	base10k	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	soft10k	0.194	0.628	4.32e+07	2.41e+08	0.616	3.99e+10	1.49e+05	1	1	0	Complete
4 T/P DeepONet10	long	0.113	0.744	7.35e+04	3.8	0.715	4.46e+10	2.45e+05	1	1	0	Complete
5 ANN10 + T/P DeepONet10	base10k	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.266	0.878	2.05e+07	6.45e+07	0.874	5.69e+10	1.85e+05	2.78e+03	1.19e+04	0	Complete
5 ANN10 + T/P DeepONet10	long	0.198	0.929	5.36e+04	53.3	0.973	5.34e+10	1	1.67e+03	5.33e+05	0	Complete
6 T/P/ϕ DeepONet10	base10k	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.26	0.575	4.98e+08	6.22e+08	0.435	2.23e+10	3.39e+05	1	1	0	Complete
6 T/P/ϕ DeepONet10	long	0.119	0.838	4.46e+05	9.03e+03	0.802	4.99e+10	2.87e+05	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.422	0.865	8.71e+06	3.21e+08	0.904	6.19e+10	3.71e+03	3.22e+03	6.97e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.207	0.925	2.08e+06	7.69e+04	0.968	5.37e+10	1	1.78e+03	5.09e+05	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 73

T_0 = 1106.0 K P_0 = 1.911 atm ϕ_0 = 1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 73 | Cantera vs NN NRMSE

T0=1106.0 K P0=1.911 atm phi0=1.915 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0381	0.0487	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	soft10k	0.682	0.636	1	1	3.2	54.2	1	1.03	59.3	0	Complete
1 ANN10	long	0.686	0.596	1	1	1.31	55.4	1	7.28	7.3e+03	0	Complete
2 P-only DeepONet10	base10k	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	soft10k	0.641	0.871	1	1	7.94	38.9	1	1	1.02	0	Complete
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.140 ms
3 ANN10 + P-only DeepONet10	base10k	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.679	0.66	1	1	3.86	53.1	1	1.02	16.4	0	Complete
3 ANN10 + P-only DeepONet10	long	0.684	0.618	1	1	2.38	54.5	1	1	96.2	0	Complete
4 T/P DeepONet10	base10k	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	soft10k	0.714	0.666	0.53	37.1	2.86	48	1	1.05	1	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.218 ms
5 ANN10 + T/P DeepONet10	base10k	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.367	0.572	1.45	26.7	5.57	1	0.585	1.32e+03	4.76e+04	0	Complete
5 ANN10 + T/P DeepONet10	long	0.687	0.588	1	1.01	0.77	55.8	1	6.13	5.34e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.749	0.606	2.14	29.2	3.63	42.3	0.956	1.12	1	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.216 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.382	0.563	2	29.7	5.77	1	0.573	1.2e+03	3.50e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.688	0.587	1	1.25	0.81	56.1	1	3.85	588	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 74

T_0 = 1222.9 K P_0 = 4.971 atm ϕ_0 = 1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference · Baseline10k (lambda=0) · Soft10k (lambda=100) · Soft long (lambda=100)

1 ANN10

2 P-only DeepONet10

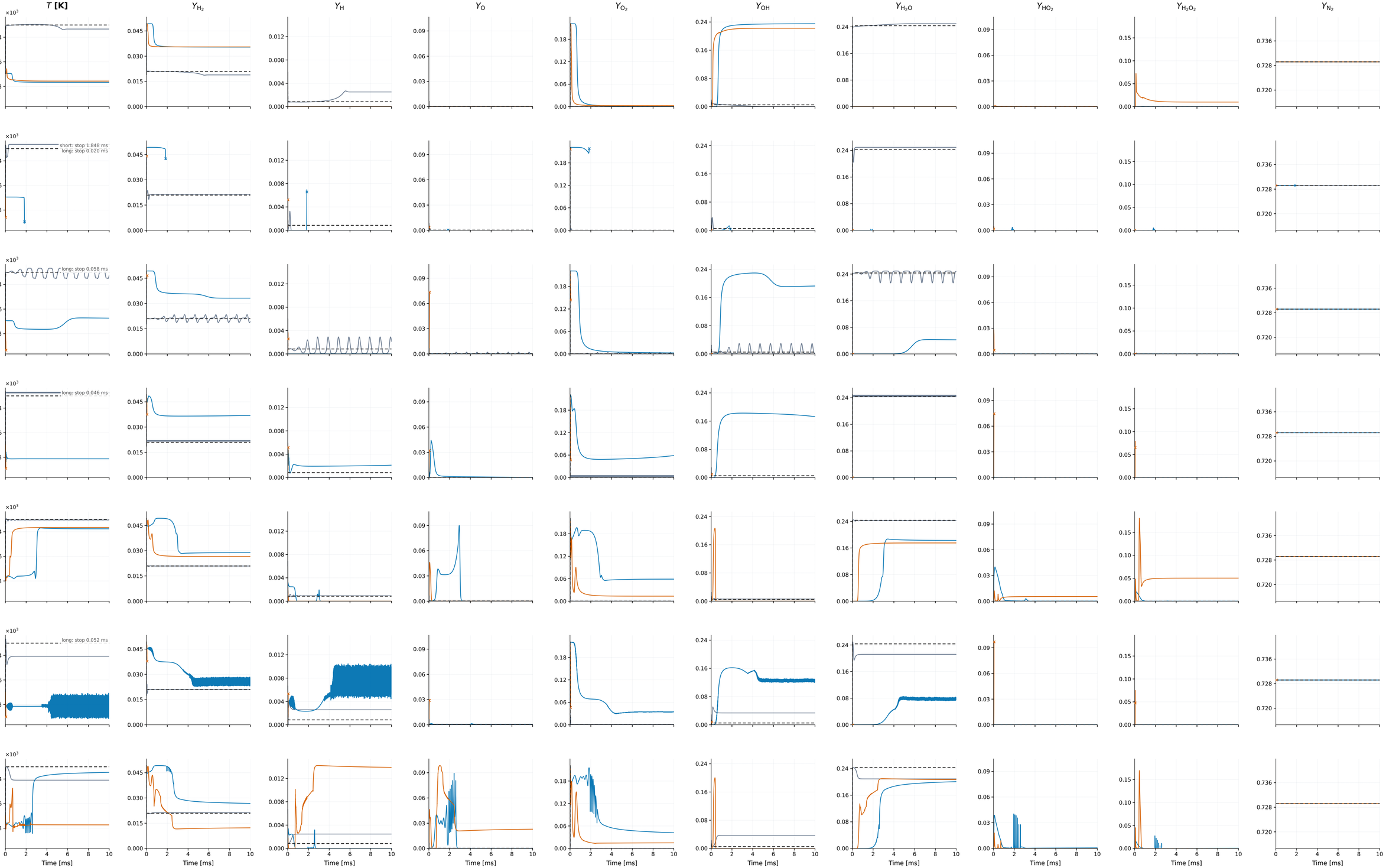
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.
Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 74 | Cantera vs NN NRMSE

T0=1222.9 K P0=4.971 atm phi0=1.766 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

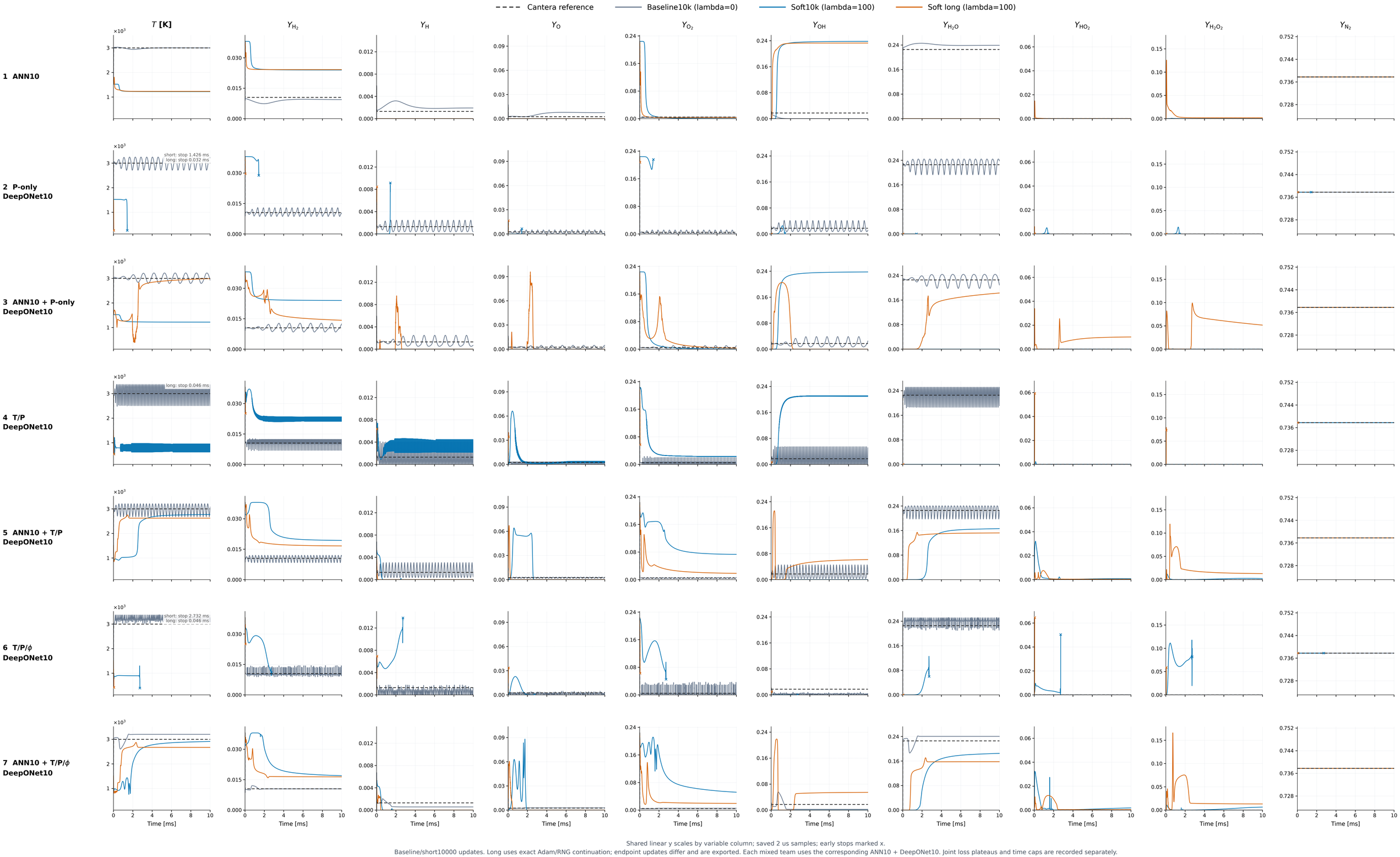
Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	soft10k	0.658	0.758	1	1	6.85	41.9	1	1.46	135	0	Complete
1 ANN10	long	0.648	0.71	1	1	2.93	40.8	1	30.4	1.79e+04	0	Complete
2 P-only DeepONet10	base10k	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.848 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
3 ANN10 + P-only DeepONet10	base10k	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.599	0.744	1	1	7.85	37.2	0.927	1.36	33.7	0	Complete
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	base10k	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	soft10k	0.731	0.791	1.37	25.5	8.37	32.1	1	6.45	2.11	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
5 ANN10 + T/P DeepONet10	base10k	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.37	0.752	1.07	74.1	12.9	1	0.564	914	4.64e+03	0	Complete
5 ANN10 + T/P DeepONet10	long	0.172	0.361	0.992	18.3	3.48	5.39	0.365	603	7.12e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.736	0.549	6.4	1.15	8.71	24.5	0.811	7.76	3.08	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.052 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.345	0.709	1.06	64.9	12.2	1	0.53	820	3.44e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.669	0.48	13.8	131	4.35	4.95	0.329	134	2.20e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 76

T_0 = 1523.6 K P_0 = 3.626 atm ϕ_0 = 1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$

Cantera reference Baseline10k (lambda=0) Soft10k (lambda=100) Soft long (lambda=100)



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 76 | Cantera vs NN NRMSE

T0=1523.6 K P0=3.626 atm phi0=1.349 | Repaired self-fed,10ms

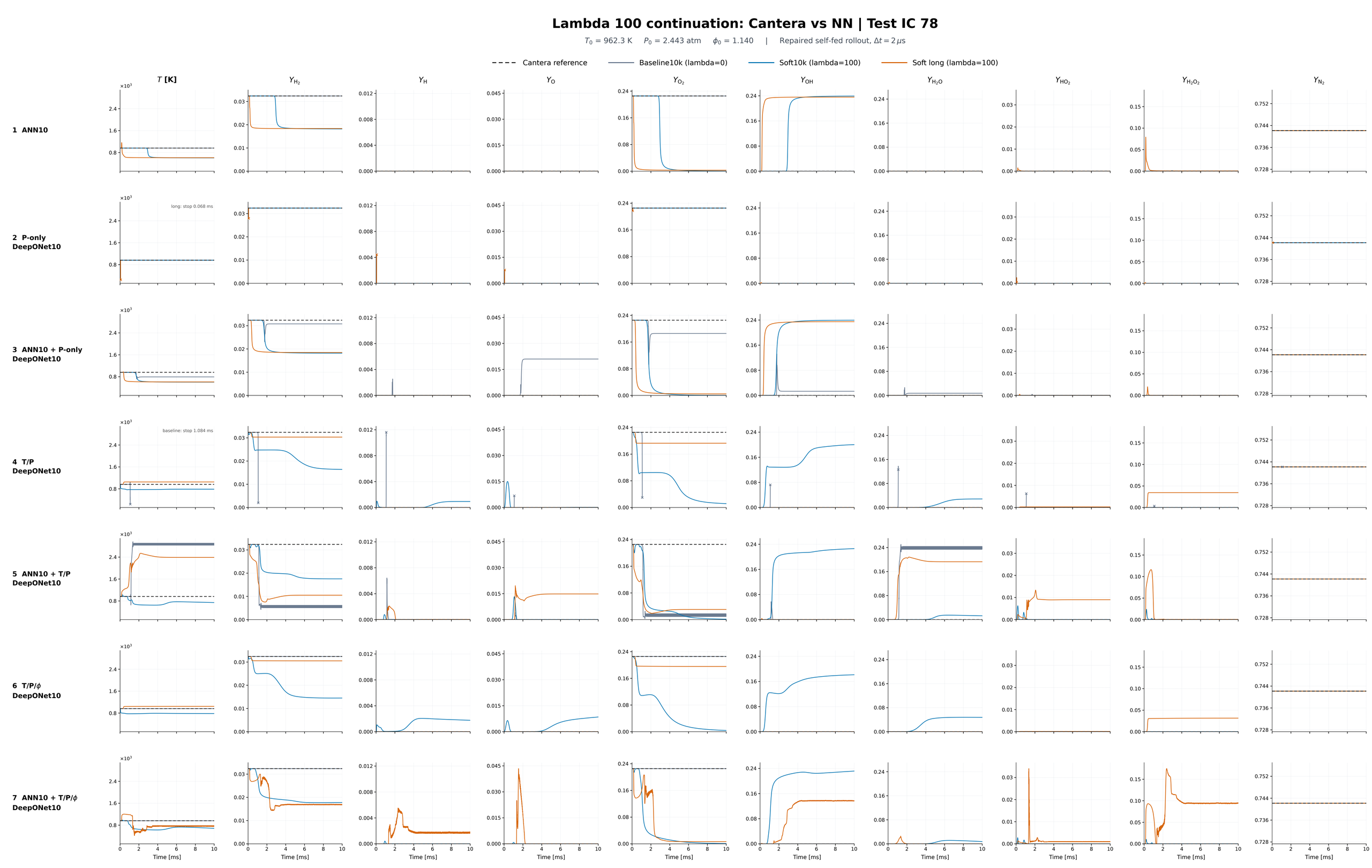
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	soft10k	0.587	1.41	1	1	8.01	12	1	1.6	164	0	Complete
1 ANN10	long	0.586	1.32	1	1	2.48	12.1	1	117	1.31e+04	0	Complete
2 P-only DeepONet10	base10k	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.426 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.032 ms
3 ANN10 + P-only DeepONet10	base10k	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.584	1.44	1	1	8.95	11.8	1	3.47	139	0	Complete
3 ANN10 + P-only DeepONet10	long	0.307	0.919	1.29	5.86	6.46	4	0.526	1.26e+03	6.72e+04	0	Complete
4 T/P DeepONet10	base10k	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	soft10k	0.736	1.28	1.71	5.25	8.01	10.4	1	27.6	20	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
5 ANN10 + T/P DeepONet10	base10k	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.342	1.49	1.08	9.6	16.2	1	0.561	829	2.88e+03	0	Complete
5 ANN10 + T/P DeepONet10	long	0.182	0.783	0.998	2.75	5.09	2.59	0.408	281	3.35e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 2.732 ms
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.046 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.287	1.32	1.07	6.63	15.2	1	0.497	827	4.32e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.206	0.765	0.991	3	4.78	2.83	0.417	599	3.99e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 78

T_0 = 962.3 K P_0 = 2.443 atm ϕ_0 = 1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 78 | Cantera vs NN NRMSE

T0=962.3 K P0=2.443 atm phi0=1.140 | Repaired self-fed,10ms

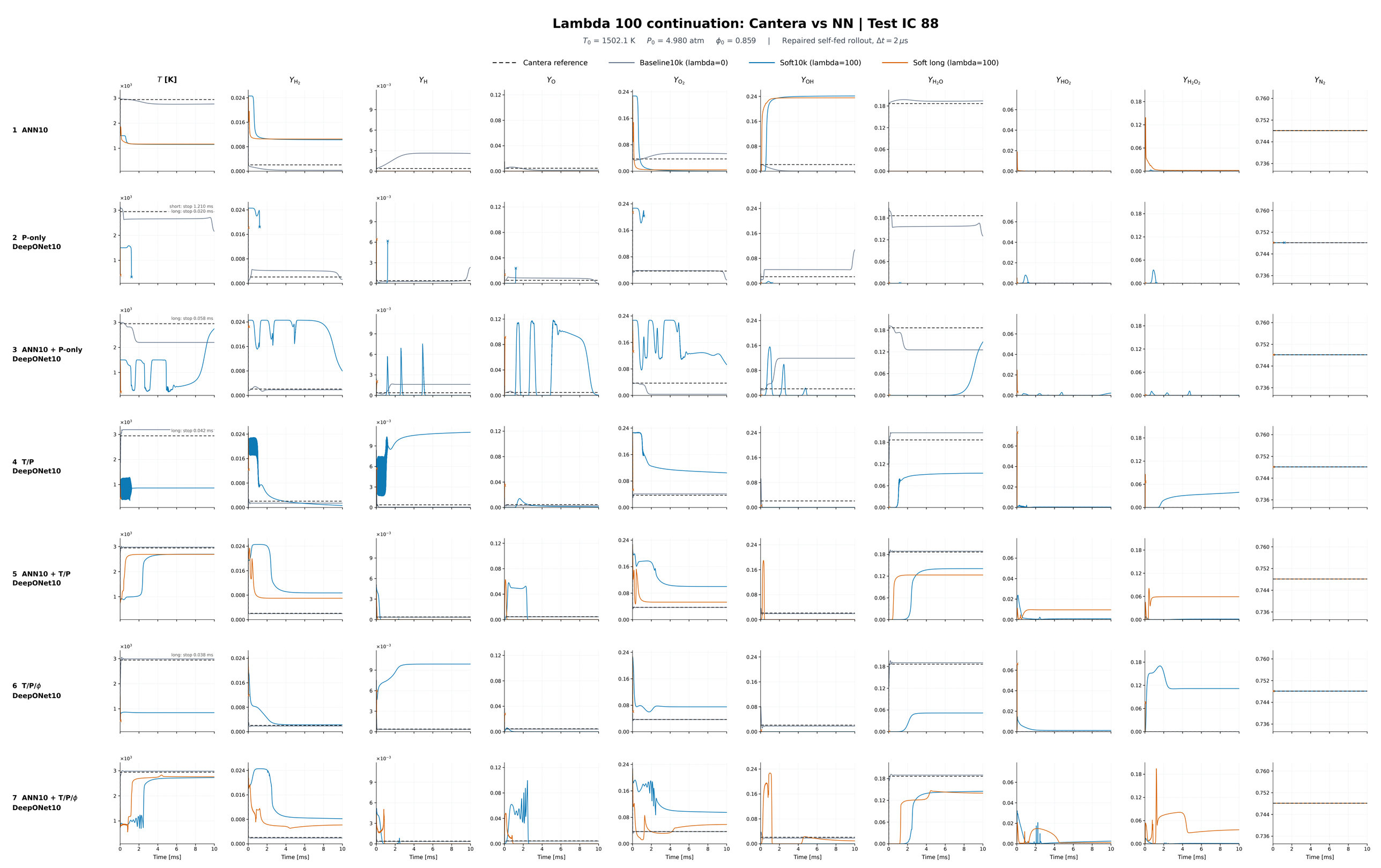
$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	soft10k	0.307	0.361	1.01	1.03	0.824	1.37e+09	1	1.93	34.6	0	Complete
1 ANN10	long	0.358	0.427	1.03	1	0.973	1.61e+09	1	26.5	1.22e+03	0	Complete
2 P-only DeepONet10	base10k	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	soft10k	5.31e-05	2.47e-05	2.04	3.62e+03	4.84e-05	1	0.912	1	1	0	Complete
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.068 ms
3 ANN10 + P-only DeepONet10	base10k	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.331	0.39	1	1	0.889	1.48e+09	1	1.69	7.98	0	Complete
3 ANN10 + P-only DeepONet10	long	0.353	0.417	4.65	1.44	0.95	1.58e+09	1	8.33	318	0	Complete
4 T/P DeepONet10	base10k	—	—	—	—	—	—	—	—	—	—	Stop 1.084 ms
4 T/P DeepONet10	soft10k	0.186	0.366	2.94e+06	9.04e+06	0.741	1.13e+09	4.68e+03	1	1	0	Complete
4 T/P DeepONet10	long	0.0915	0.0622	1	1	0.142	568	1	55.8	7.28e+03	0	Complete
5 ANN10 + T/P DeepONet10	base10k	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.244	0.396	4.90e+05	6.27e+06	0.87	1.39e+09	2.62e+03	153	484	0	Complete
5 ANN10 + T/P DeepONet10	long	1.41	0.653	2.56e+06	4.73e+07	0.833	1.85e+06	4.92e+04	1.82e+03	6.13e+03	0	Complete
6 T/P/ϕ DeepONet10	base10k	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.184	0.452	8.08e+06	1.75e+07	0.806	1.08e+09	9.86e+03	1	1	0	Complete
6 T/P/ϕ DeepONet10	long	0.083	0.0563	1	1.07	0.129	578	0.972	20.9	6.6e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	base10k	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.277	0.404	2.30e+05	3.06e+05	0.898	1.45e+09	1.87e+03	64.8	203	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.242	0.432	1.08e+07	2.60e+07	0.858	7.92e+08	1.02e+03	488	2.04e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.

Lambda 100 continuation: Cantera vs NN | Test IC 88

T_0 = 1502.1 K P_0 = 4.980 atm ϕ_0 = 0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu s$



Shared linear y scales by variable column; saved 2 us samples; early stops marked x.

Baseline/short10000 updates. Long uses exact Adam/RNG continuation; endpoint updates differ and are exported. Each mixed team uses the corresponding ANN10 + DeepONet10. Joint loss plateaus and time caps are recorded separately.

Lambda 100 | Test IC 88 | Cantera vs NN NRMSE

T0=1502.1 K P0=4.980 atm phi0=0.859 | Repaired self-fed,10ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	base10k	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	soft10k	0.6	4.64	1	1	1.52	9.93	1	1.06	196	0	Complete
1 ANN10	long	0.599	4.08	1	1	0.928	9.98	1	34.3	5.43e+03	0	Complete
2 P-only DeepONet10	base10k	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	soft10k	—	—	—	—	—	—	—	—	—	—	Stop 1.210 ms
2 P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.020 ms
3 ANN10 + P-only DeepONet10	base10k	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	soft10k	0.674	9.55	2.59	14	3.35	1.55	0.936	27.4	1.04e+03	0	Complete
3 ANN10 + P-only DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.058 ms
4 T/P DeepONet10	base10k	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	soft10k	0.709	2.81	24.8	0.764	2.57	1	0.58	19.8	1.41e+04	0	Complete
4 T/P DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.042 ms
5 ANN10 + T/P DeepONet10	base10k	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	soft10k	0.334	5.76	1.79	4.57	2.4	1	0.536	134	683	0	Complete
5 ANN10 + T/P DeepONet10	long	0.159	2.91	1.07	1.72	0.718	1.39	0.402	332	2.74e+04	0	Complete
6 T/P/ϕ DeepONet10	base10k	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	soft10k	0.712	1.49	22.9	0.968	1.14	1	0.788	93.8	5.73e+04	0	Complete
6 T/P/ϕ DeepONet10	long	—	—	—	—	—	—	—	—	—	—	Stop 0.038 ms
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.346	5.66	2.16	5.02	2.32	1	0.535	203	973	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	long	0.246	2.53	1.76	1.66	0.592	2.58	0.442	253	2.47e+04	0	Complete

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Each graph row includes Cantera, baseline10k, same-lambda soft10k and soft-long.